Zilong Zhang

Ph.D. Student in Statistics and Machine Learning, Georgia State University
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EDUCATION

Ph.D. in Machine Learning & Data Science **Aug. 2025 – Present**
Department of Mathematics & Statistics / Georgia State University *Atlanta, GA*

M.S. in Statistics **Aug. 2023 – May 2025**
Department of Statistics & Probability / Michigan State University *East Lansing, MI*

Bachelor of Financial Mathematics **Sep. 2019 – Jul. 2023**
School of Statistics / University of International Business and Economics *Beijing, China*

RESEARCH PROFILE

- Statistics and machine-learning Ph.D. researcher studying how to make model evaluation more reliable when human labels are scarce, selective, or costly to obtain.
- Research interests include LLM-as-a-judge auditing, human-aligned evaluation, positive–unlabeled learning, uncertainty-guided review, and graph/sequence models for biomedical data.

PUBLICATIONS & MANUSCRIPTS

- **Zilong Zhang**, Yi-Ting Hung, Weiyi He, Junxi Zhang, Lei Ding, Chi-Kuang Yeh. "AURA: Adaptive Uncertainty-aware Refinement for LLM-as-a-Judge Auditing" *Under review, arXiv:2606.19714*, 2026.
- **Zilong Zhang**, Yi-Ting Hung, Lei Ding, Chi-Kuang Yeh. "Quantifying and Auditing LLM Evaluation via Positive–Unlabeled Learning" *Under review, arXiv:2606.19057*, **2026**.

RESEARCH AND OTHER PROJECTS

Uncertainty-Aware Auditing of LLM-as-a-Judge Systems **2026**

- First-author research on adaptive auditing of pairwise LLM judgments: estimate alignment with human preferences, propagate reliable evidence, and prioritize uncertain comparisons for limited human review. The approach is evaluated on synthetic and real pairwise-response data.
- Complementary first-author work frames selectively human-labeled comparisons as positive–unlabeled learning and uses partial optimal transport to align verified positives with a reliable subset of unlabeled examples. The goal is to improve human-preference alignment and robustness to judge/verbosity bias without retraining the judge.

Functional Time-Series Research **Sept. 2025 – Present**

- Research on bandwidth selection for functional time series and long-run covariance estimation.

mRNA-to-Protein Prediction with Transformers and GNNs **Jan. 2024 – Jan. 2026**

- Built a Transformer/GNN model to predict protein expression from single-cell RNA data, using complementary sequence/context and graph representations.
- Conducted the cross-institution project with collaborators at Michigan State University and MD Anderson Cancer Center, connecting single-cell RNA inputs to protein-level outcomes.

WORK EXPERIENCE

China Securities Co., Ltd. **Jul. 2021 – Nov. 2021**
Investment Management Intern *Beijing, China*

- Prepared industry research, risk assessments, listing materials, and fundraising reports for client companies.

TECHNICAL SKILLS

Programming: Python, R, C; PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
Research methods: LLM evaluation, positive-unlabeled learning, partial optimal transport, uncertainty estimation, GNNs
Tools: Git, Linux, Jupyter, VS Code